

## 5.5.1 Substitution w/ Indefinite Integrals Worksheet

Use a u-substitution to evaluate each of the following.

1.  $\int x \sin(x^2) dx$

Let  $u = x^2$ , then  $du = 2x dx \rightarrow \frac{1}{2} du = x dx$

Therefore,

$$\int x \sin(x^2) dx = \int \frac{1}{2} \sin(u) du = -\frac{1}{2} \cos(u) + C = -\frac{1}{2} \cos(x^2) + C$$

2.  $\int (1-2x)^9 dx$

Let  $u = 1-2x$ , then  $du = -2dx \rightarrow -\frac{1}{2} du = dx$

Therefore,

$$\int (1-2x)^9 dx = \int -\frac{1}{2} (u)^9 du = -\frac{1}{20} u^{10} + C = -\frac{1}{20} (1-2x)^{10} + C$$

3.  $\int \sec(3t) \tan(3t) dt$

Let  $u = 3t$ , then  $du = 3dt \rightarrow \frac{1}{3} du = dt$

Therefore,

$$\int \sec(3t) \tan(3t) dt = \int \frac{1}{3} \sec(u) \tan(u) du = \frac{1}{3} \sec(u) + C = \frac{1}{3} \sec(3t) + C$$

4.  $\int (x+1)\sqrt{2x+x^2} dx$

Let  $u = 2x+x^2$ , then  $du = (2+2x)dx \rightarrow \frac{1}{2} du = (1+x)dx$

Therefore,

$$\int (x+1)\sqrt{2x+x^2} dx = \int \frac{1}{2} \sqrt{u} du = \int \frac{1}{2} u^{1/2} du = \frac{1}{3} u^{3/2} + C = \frac{1}{3} (2x+x^2)^{3/2} + C$$

5.  $\int \sec^2(\theta) \tan^3(\theta) d\theta$

Let  $u = \tan(\theta)$ , then  $du = \sec^2(\theta) d\theta$

Therefore,

$$\int \sec^2(\theta) \tan^3(\theta) d\theta = \int u^3 du = \frac{1}{4}u^4 + C = \frac{1}{4}\tan^4(\theta) + C$$

6.  $\int \frac{\cos(x)}{\sin^2(x)} dx$

Let  $u = \sin(x)$ , then  $du = \cos(x) dx$

Therefore,

$$\int \frac{\cos(x)}{\sin^2(x)} dx = \int \frac{1}{u^2} du = \int u^{-2} du = -u^{-1} + C = -(\sin(x))^{-1} + C = -\frac{1}{\sin(x)} + C = -\csc(x) + C$$

7.  $\int \frac{z^2}{\sqrt[3]{1+z^3}} dz$

Let  $u = 1 + z^3$ , then  $du = 3z^2 dz \rightarrow \frac{1}{3} du = z^2 dz$

Therefore,

$$\int \frac{z^2}{\sqrt[3]{1+z^3}} dz = \int \frac{1}{3} \frac{1}{\sqrt[3]{u}} du = \int \frac{1}{3} u^{-1/3} du = \frac{1}{2} u^{2/3} + C = \frac{1}{2} (1+z^3)^{2/3} + C$$

8.  $\int (x^2 + 1)(x^3 + 3x)^4 dx$

Let  $u = x^3 + 3x$ , then  $du = (3x^2 + 3) dx \rightarrow \frac{1}{3} du = (x^2 + 1) dx$

Therefore,

$$\int (x^2 + 1)(x^3 + 3x)^4 dx = \int \frac{1}{3} (u)^4 du = \frac{1}{12} u^5 + C = \frac{1}{12} (x^3 + 3x)^5 + C$$

9.  $\int \sqrt{\cot(x)} \csc^2(x) dx$

Let  $u = \cot(x)$ , then  $du = -\csc^2(x) dx \rightarrow -du = \csc^2(x) dx$

Therefore,

$$\int \sqrt{\cot(x)} \csc^2(x) dx = \int -\sqrt{u} du = \int -u^{1/2} du = -\frac{2}{3} u^{3/2} + C = -\frac{2}{3} (\cot(x))^{3/2} + C$$

10.  $\int \cos^4(x) \sin(x) dx$

Let  $u = \cos(x)$ , then  $du = -\sin(x) dx \rightarrow -du = \sin(x) dx$

Therefore,

$$\int \cos^4(x) \sin(x) dx = \int -u^4 du = \int -\frac{1}{5} u^5 du = -\frac{1}{30} u^6 + C = -\frac{1}{30} \cos^6(x) + C$$

11.  $\int x(2x+5)^8 dx$

Let  $u = 2x+5$ , then  $du = 2dx \rightarrow \frac{1}{2} du = dx$ . We also know that  $x = \frac{1}{2}(u-5)$ .

Therefore,

$$\int x(2x+5)^8 dx = \int \frac{1}{2}(u-5)u^8 du = \int \frac{1}{2}(u^9 - 5u^8) du = \frac{1}{2} \left( \frac{1}{10} u^{10} - \frac{5}{9} u^9 \right) + C = \frac{1}{20} (2x+5)^{10} - \frac{5}{90} (2x+5)^9 + C$$

12.  $\int \frac{\cos(\pi/x)}{x^2} dx$

Let  $u = \frac{\pi}{x}$ , then  $du = -\frac{\pi}{x^2} dx \rightarrow -\frac{1}{\pi} du = \frac{1}{x^2} dx$

Therefore,

$$\int \frac{\cos(\pi/x)}{x^2} dx = \int -\frac{1}{\pi} \cos(u) du = -\frac{1}{\pi} \sin(u) + C = -\frac{1}{\pi} \sin\left(\frac{\pi}{x}\right) + C$$

13.  $\int x^2 \sqrt{2+x} dx$

Let  $u = 2 + x$ , then  $du = dx$ . We also know that  $x = u - 2$ .  
Therefore,

$$\begin{aligned} \int x^2 \sqrt{2+x} dx &= \int (u-2)^2 \sqrt{u} du = \int (u^2 - 4u + 4) u^{1/2} du = \int (u^{5/2} - 4u^{3/2} + 4u^{1/2}) du = \frac{2}{7} u^{7/2} - \frac{8}{5} u^{5/2} + \frac{8}{3} u^{3/2} + C \\ &= \frac{2}{7} (2+x)^{7/2} - \frac{8}{5} (2+x)^{5/2} + \frac{8}{3} (2+x)^{3/2} + C \end{aligned}$$

14.  $\int \sin(t) \sec^2(\cos(t)) dt$

Let  $u = \cos(t)$ , then  $du = -\sin(t) dt \rightarrow -du = \sin(t) dt$

Therefore,

$$\int \sin(t) \sec^2(\cos(t)) dt = \int -\sec^2(u) du = -\tan(u) + C = -\tan(\cos(t)) + C$$

15.  $\int x^3 \sqrt{x^2+1} dx$

Let  $u = x^2 + 1$ , then  $du = 2x dx \rightarrow \frac{1}{2} du = x dx$ . We also know that  $x^2 = u - 1$ .

Therefore,

$$\begin{aligned} \int x^3 \sqrt{x^2+1} dx &= \int \frac{1}{2} (u-1) \sqrt{u} du = \int \frac{1}{2} (u^{3/2} - u^{1/2}) du = \frac{1}{2} \left( \frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right) + C = \frac{1}{5} u^{5/2} - \frac{1}{3} u^{3/2} + C \\ &= \frac{1}{5} (x^2+1)^{5/2} - \frac{1}{3} (x^2+1)^{3/2} + C \end{aligned}$$

16.  $\int \frac{dx}{5-3x}$

Let  $u = 5 - 3x$ , then  $du = -3dx \rightarrow -\frac{1}{3} du = dx$

Therefore,

$$\int \frac{dx}{5-3x} = \int -\frac{1}{3} \frac{1}{u} du = -\frac{1}{3} \ln|u| + C = -\frac{1}{3} \ln|5-3x| + C$$

17.  $\int e^{\tan(x)} \sec^2(x) dx$

Let  $u = \tan(x)$ , then  $du = \sec^2(x) dx$

Therefore,

$$\int e^{\tan(x)} \sec^2(x) dx = \int e^u du = e^u + C = e^{\tan(x)} + C$$

18.  $\int \cot(x) dx$

First we note that  $\int \frac{\cos(x)}{\sin(x)} dx$ .

Let  $u = \sin(x)$ , then  $du = \cos(x) dx$

Therefore,

$$\int \frac{\cos(x)}{\sin(x)} dx = \int \frac{1}{u} du = \ln|u| + C = \ln|\sin(x)| + C$$

19.  $\int e^x \sqrt{1+e^x} dx$

Let  $u = 1 + e^x$ , then  $du = e^x dx$ .

Therefore,

$$\int e^x \sqrt{1+e^x} dx = \int \sqrt{u} du = \frac{2}{3} u^{3/2} + C = \frac{2}{3} (1+e^x)^{3/2} + C$$

20.  $\int \frac{\sin(x)}{1+\cos^2(x)} dx$

Let  $u = \cos(x)$ , then  $du = -\sin(x) dx \rightarrow -du = \sin(x) dx$ .

Therefore,

$$\int \frac{\sin(x)}{1+\cos^2(x)} dx = \int \frac{-1}{1+u^2} du = -\arctan(u) + C = -\arctan(\cos(x)) + C$$

21.  $\int \frac{1+x}{1+x^2} dx$  (Consider using some algebra first before attempting a substitution)

First, we note that

$$\int \frac{1+x}{1+x^2} dx = \int \left( \frac{1}{1+x^2} + \frac{x}{1+x^2} \right) dx = \int \frac{1}{1+x^2} dx + \int \frac{x}{1+x^2} dx = \arctan(x) + C + \int \frac{x}{1+x^2} dx$$

On the remaining integral we let  $u = 1+x^2$ , then  $du = 2x dx \rightarrow \frac{1}{2} du = x dx$ .

$$\text{Therefore, } \int \frac{x}{1+x^2} dx = \int \frac{1}{2} \frac{1}{u} du = \frac{1}{2} \ln|u| + C = \frac{1}{2} \ln|1+x^2| + C$$

Overall, this gives us that

$$\int \frac{1+x}{1+x^2} dx = \arctan(x) + \frac{1}{2} \ln|1+x^2| + C$$